

Mapping Cerebello-Cortical Structural Connectivity Associated with Cognitive Impairment in Essential Tremor

Sajan C. Patel

MEng. Biomedical Engineering

Duke University

Durham, N.C., U.S.A

sajan.patel@duke.edu

April 28, 2026

Abstract—Essential Tremor (ET) is historically characterized as a motor control disorder, but recent connectomic evidence links Lobule VIIIB of the cerebellum to complex cognitive deficits. To investigate the white matter tracts underlying cognitive versus motor function, this study introduces an automated Python-based pipeline for computational tractography and coordinate network mapping. The pipeline was utilized to assess the structural connectivity of cerebellar Lobules V and VIIIB to predefined cerebral motor and non-motor (cognitive, sensory, and limbic) networks derived from Peak Voxel Coordinates. Network segregation was quantitatively evaluated using a Chi-Squared Test of Independence and Cramer’s V effect size. The analysis revealed that both Lobules V and VIIIB predominantly intersect with the motor control network. No statistically significant double dissociation was observed between the anterior and posterior lobules regarding their motor versus cognitive connectivity profiles in either hemisphere. Furthermore, a brain-wide 34x34 pairwise comparison matrix demonstrated widespread structural homogeneity across the cerebellum, though isolated pairs such as the Right Dentate and Left Lobule IX exhibited moderate functional segregation. While limited by deterministic tractography constraints and alignment precision, these findings emphasize the necessity for continued, high-resolution exploration into the structural basis of the cerebellum’s distinct functional specializations.

Keywords—Essential Tremor, Structural Connectivity, Computational Tractography, Connectomics, Cerebellum

I. INTRODUCTION

A. Background & Research Opportunities.

Essential Tremor has traditionally been viewed as a motor control disorder because its defining characteristic is abnormal oscillatory movement that produces both action and postural tremor [1]. However, recent connectomic mapping has demonstrated that the cerebellum acts as a central hub with distinct functional specializations across motor, cognitive, sensory, and limbic networks [2]. Though cerebellar Lobules I-VI have primarily been associated with motor tremor, Lobule VIIIB has recently been linked to complex cognitive deficits that are often observed in ET patients [3]. These include impairment of executive function, working memory, verbal fluency, and visuospatial processing [4]. White matter tracts linking specific

cerebellar subregions – such as Lobule VIIIB – to the cerebral cortex remain poorly characterized. This provides a clear opportunity to investigate their structural connectivity across various functional networks.

B. Literature Review

Since ET is generally not caused by standard lesions (such as from a stroke or tumor), previous research has utilized “coordinate network mapping” (CNM) to identify common functional networks [3]. This involves seeding many Peak Voxel Coordinates (PVCs) into a RS-fMRI connectome, where each PVC is the specific spatial point within a cluster that had the highest statistical significance (t-value or z-score) for a given functional or structural alteration [3]. From this CNM work, the cerebellum was the only region functionally connected to the findings of over 90% of the studies the authors of *Younger et al. (2023)* looked at [3]. They verified that connectivity between the cerebellum and sensorimotor cortices was significantly reduced compared to healthy controls, and that this network correlated with severity of tremors and cognitive function in ET patients [3].

An earlier paper, *Fang et al. (2013)*, had found a variety of PVCs in the cerebello-thalamo-cortical motor pathway that correlated with generation and propagation of tremors in ET patients [2]. Here, they used Regional Homogeneity (ReHo) as the metric to assess specific points of functional connectivity abnormalities through RS-fMRI [2]. Other PVCs found in the prefrontal and parietal cortices may explain the presence of non-motor symptoms sometimes found in ET patients [2].

C. Project Overview

This project introduces an automated, highly-customizable Python-based approach for warping a cerebellar atlas into patient space, performing computational tractography, and constructing functional networks via PVCs, alongside integrated statistical analysis, and dynamic visualization. This enables researchers to analyze connections between any cerebellar subregions and several functional whole-brain networks. Additionally, dynamic 3D visualizations produced by the pipeline can enable neuroanatomists to more easily

assess if these *in silico* tractography reconstructions match true human neuroanatomy.

Specifically, this pipeline is applied to connectivity analysis of cerebellar Lobules V and VIIB with several functional networks, such as motor control, cognitive, sensory, and limbic. An improved understanding of these circuits could guide future Deep-Brain Stimulation (DBS) to counter cognitive deficits in ET or to avoid stimulation of these specific subregions in Parkinson's patients to prevent meddling with cognitive processes.

II. METHODS

A. Peak Voxel Coordinates Collection

First, a small database of PVCs were collected from various papers and sorted into four distinct functional networks: Motor Control (13 coordinates), Prefrontal Cortex & Executive Control (10 coordinates), Medial Dorsal Thalamus (5 coordinates), Parietal/Precuneus/ Default Mode Network (8 coordinates) [9].

B. Preprocessing Steps

The single-subject T1-weighted (T1w) anatomical MRI scan and 4D diffusion-weighted image (DWI) given in the BME 502 course homework were collected. The T1w scan was passed through BrainSuite's Surface-Volume Registration (SVReg) pipeline [5]. This included Atlas Selection (USCBrain), Cortical Thickness Estimation, Automated Labeling, Refinement, and Volumetric Statistics. Next, the DWI was passed through BrainSuite's Diffusion Pipeline (BDP) to prepare the data for connectivity analysis [5]. This included Distortion Correction, Diffusion Modeling, and ODF Estimation.

C. Constructing Custom Python Pipeline

To construct a custom pipeline, the NiBabel, NiLearn, and DIPY Python packages were utilized heavily (with assistance from Google Gemini AI) [6]-[8]. First, linear transformation was performed to warp the SUIT Cerebellar Atlas to the T1w anatomical MRI scan using the bias-field corrector and atlas in MNI space [10]-[14]. Next, streamlines were constructed for the entire brain using the 4D diffusion image, b-values, and rotated b-vectors from the BDP.

Each PVC was expanded into a 5 mm sphere-of-interest (SOI) to represent a small volume of tissue, rather than a single point. These SOIs were projected into the same 3D space as the streamlines. To calculate the connectivity of a given cerebellar subregion to a given functional network, the set of all streamlines intersecting with the subregion were separated from the rest of the brain's streamlines. This set was then separated into two subsets: 1) intersect both cerebellar subregion and a SOI in the functional network; 2) intersect cerebellar subregion only. This step was repeated for all possible pairs of a cerebellar subregion (34 total) and a functional network (4 total).

D. Statistical Analysis

To evaluate the double dissociation of structural connectivity between motor and cognitive networks across the cerebellum, a series of χ^2 (Chi-Squared) Tests of Independence

were conducted. The primary objective was to determine whether the distribution of streamlines intersecting with specific cerebral functional networks is independent of the cerebellar lobules they also intersect with.

Target subregions-of-interest were grouped into two broad categories: "Motor Network" (Peak Voxel Coordinates of motor control) and "Cognitive Network" (Peak Voxel Coordinates from the Prefrontal Cortex, Medial Dorsal Thalamus, and Parietal Lobes/Precuneus/Default Mode Network). For each pairwise subregional comparison (e.g., Left Lobule V vs. Left Lobule VIIB), a 2x2 contingency table was constructed with raw streamline counts (rows = cerebellar subregion, columns = functional network). The test statistic was calculated using the standard formula:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

where "O" is the observed streamline count, "E" is the expected streamline count under the null hypothesis of independence, and α (statistical significance) was set at $p < 0.05$.

Since tractography generates tens of thousands of streamlines, traditional null-hypothesis is very susceptible to Type I errors. To account for this and measure the actual magnitude of network segregation, Cramer's V (V) was calculated for every comparison as a measure of effect size:

$$V = \sqrt{\frac{\chi^2}{N \min(r-1, c-1)}}$$

where N is the total number of streamlines connecting the two subregions to the two networks, r is the number of rows, and c is the number of columns.

Then, a brain-wide network segregation analysis was performed. The 2x2 χ^2 test and Cramer's V calculation were iterated across all possible unique pairs of the 34 cerebellar atlas subregions. This produced a 34x34 heatmap matrix to map motor-cognitive functional specialization across the cerebellum.

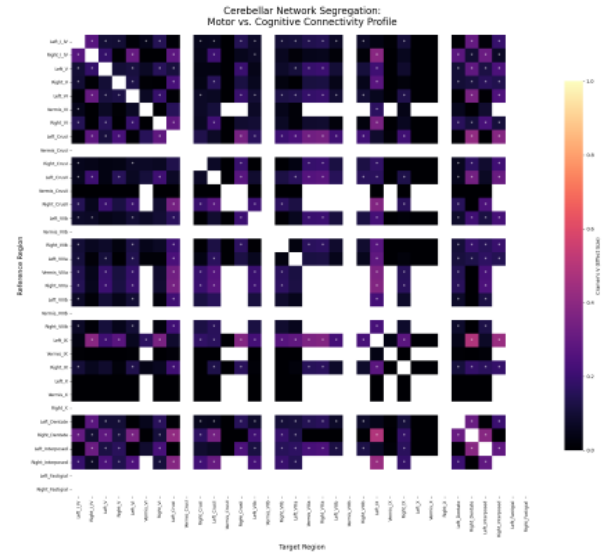


Fig 1. 34x34 pairwise comparison matrix (χ^2) was generated to assess connectivity to Motor and Non-Motor Networks across the entire SUIT Cerebellar atlas.

III. RESULTS

A. Overall Connectivity Topography

The Cognitive, Sensory, and Limbic networks combined will be referred to as “Non-Motor” networks across both Cerebellar Lobule V and VIIB. In the left hemisphere, Lobule V produced 1566 total streamlines, with 119 intersecting with the Motor Control network and 18 intersecting with the Non-Motor networks. Lobule VIIB produced 911 total streamlines, with 42 intersecting with the Motor Control network and 3 intersecting with the Non-Motor networks. In the right hemisphere, Lobule V produced 1219 total streamlines, with 59 intersecting with the Motor Control network and 2 intersecting with the Non-Motor networks. Lobule VIIB produced 870 total streamlines, with 87 intersecting with the Motor Control network and 4 intersecting with the Non-Motor networks [Fig. 2].

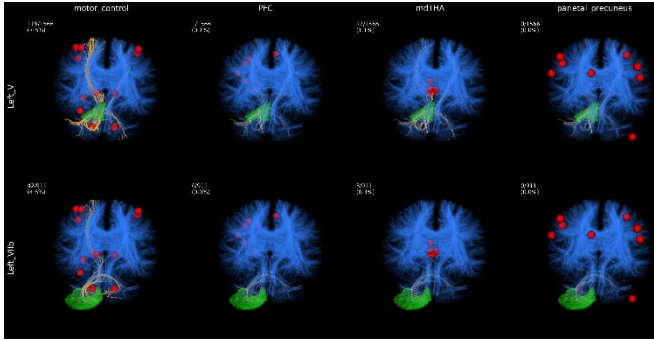


Fig 2. Visualization of subregion-network intersection for left-hemisphere cerebellar Lobules V and VIIB.

B. Two-Lobule Segregation Analysis

Hypothesis: For streamlines that intersect both a Functional Network and either Cerebellar Lobule V or VIIB, the Functional Network that a streamline intersects with depends on which cerebellar lobule it is connected to.

To test this hypothesis of a double dissociation between Lobule V and Lobule VIIB for Motor Control network versus Non-Motor networks segregation, a χ^2 Test of Independence was performed for each hemisphere. In the left hemisphere, no statistically significant association was found between the cerebellar lobular intersection and Functional Network intersection ($\chi^2 = 0.61$, $p = 0.4337$, $df = 1$). The calculated effect size was negligible ($V = 0.0436$), indicating that the connection patterns in L. Lobule V and VIIB are almost identical.

Similarly, in the right hemisphere, no statistically significant association was found between the cerebellar lobular intersection and Functional Network intersection ($\chi^2 = 0.16$, $p = 0.6891$, $df = 1$). The calculated effect size was negligible ($V = 0.0304$), indicating that the connection patterns in R. Lobule V and VIIB are almost identical.

The data did not support the presence of a functional-structural double dissociation for Cerebellar Lobules V and VIIB in either hemisphere of this subject's connectome, since both lobules had dominant connectivity to the Motor Control network.

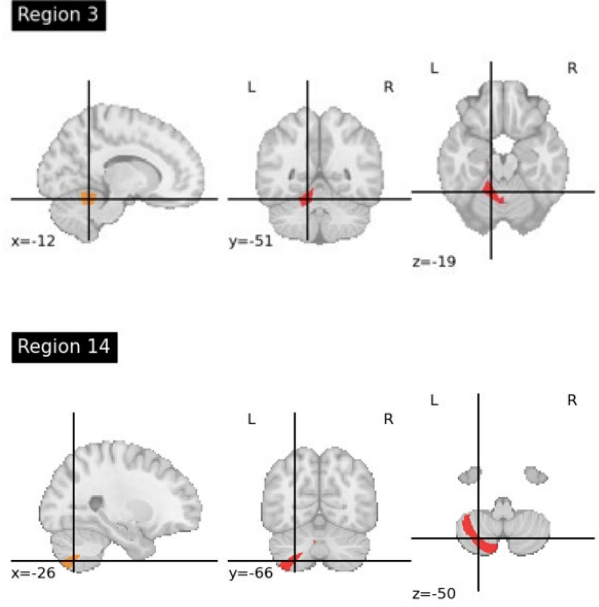


Fig 3. Left-Hemisphere Cerebellar Lobules V (top) and VIIB (bottom) from SUI Cerebellar Atlas

C. Cerebellum-Wide Topographical Segregation

A 34x34 pairwise comparison matrix was generated to assess connectivity to Motor and Non-Motor Networks across the entire SUI Cerebellar atlas [Fig. 1]. The vast majority of pairwise subregional comparisons yielded negligible effect sizes (Cramer's $V < 0.2$), displayed as dark-colored squares.

While many subregion pairs achieved statistical significance ($p < 0.05$, denoted by asterisks), the low corresponding effect sizes indicated that these differences were largely driven by massive discrepancies in sample sizes (streamline counts) that are inherent in computational tractography, rather than true biological segregation of function.

Several subregions had zero connectivity to both the motor and cognitive networks in this subject. These include a few Vermis subregions, Lobule X, and the fastigial nuclei.

A few pairwise comparisons displayed moderate network segregation. For instance, R. Denate vs. L. Crus I, R. Denate vs. Left IX, and Right IV vs. Left IX all had higher effect sizes ($V \sim 0.4 - 0.6$).

IV. DISCUSSION

Younger *et al.* (2023) found that, in ET patients, functional connectivity of Lobules VIIB/VIII to the sensorimotor cortices is associated with differences in cognitive performance, while functional connectivity of Lobule V to the sensorimotor cortices is associated with the physical tremors in ET [3]. This suggests that a separation of function exists between the anterior (Lobule V) and posterior (Lobule VIIB/VIII) cerebellar subregions. However, the results of the χ^2 Test of Independence between Cerebellar Lobule V and VIIB in both

hemispheres indicates that Lobule VIIB does not participate more in Cognitive, Sensory, and Limbic networks more than Lobule V. The tiny effect size shows that there is very little difference between Lobule V and VIIB in terms of their distribution of streamlines that intersect with the Motor Control network and other functional networks.

Another result supporting this conclusion is that there is a far greater percentage of streamlines involved in the Motor Control network compared to the Non-Motor networks for both Lobule V (7.6% and 1.1%, respectively) and VIIB (4.6% and 0.3%, respectively). Therefore, it appears that Lobule VIIB is far more involved in motor function than cognitive, sensory, and limbic function. While it may seem like this result refutes the finding of *Younger et al. (2023)* that linked Lobule VIIB to complex cognitive deficits in ET, this graduate-course project lacks the scientific rigor and verification to meaningfully disprove this cerebellar connectivity pattern [3]. However, these findings suggest that further exploration of the cerebellum's distinct functional specializations must be pursued.

While Lobule V and VIIB do not appear to have meaningful differences in functional connectivity, the 34x34 pairwise comparison matrix across all subregions of the SUIT Cerebellar Atlas points to several pairs with moderate discrepancies in functional connectivity across the Motor Control network vs. Non-Motor networks. These pairs include R. Denate vs. L. Crus I, R. Denate vs. Left IX, and Right IV vs. Left IX. A deeper look into the connectivity of these subregions would be a strong next step for a future iteration of this project. Interestingly, several subregions had zero connectivity to any of the functional networks included, such as a few Vermis lobules, Lobule X, and the fastigial nuclei. This may be evidence that these subregions participate in more localized neural circuits, rather than networks that cross the full length of the brain to connect the cerebellum to the cerebral cortex.

A. Limitations

The first notable limitation of this study is that affine/linear transformation was performed to align the SUIT Cerebellar Atlas with the T1w MRI scan. Affine/linear transformation only has 9-12 degrees of freedom, so the cerebellar subregions were projected onto the subject's scan with only rough precision. Some of the subregions look poorly aligned, which hugely impacts the streamline intersection results because the positioning of each subregion determines which streamlines are found to intersect with it.

Secondly, the pipeline only assessed whether streamlines *intersect* with a subregion and any network SOI, not whether they *originated* or *terminated* at these locations. The primary impact of a neuron's action potential is at its synapses, where neurotransmitters released there may affect nearby cells. So,

just passing through a given subregion probably does not meaningfully affect the activity of that neural tissue.

Thirdly, this approach did not distinguish streamlines that intersected *both* Lobule V and VIIB from streamlines that only intersected with one of the subregions. This means streamlines matching these criteria were double-counted because they were included in both the Lobule V and VIIB streamline counts.

Finally, only 36 PVCs were used and streamline counts were not controlled for the difference in PVCs between each network (13 in Motor Control vs. 5 in md-Thalamus). This study would be more thorough if a more comprehensive search of literature PVCs was conducted.

REFERENCES

- [1] Johns Hopkins Medicine, "Essential tremor disorder," Johns Hopkins Medicine Health Library. [Online]. Available: <https://www.hopkinsmedicine.org/health/conditions-and-diseases/essential-tremor-disorder>. [Accessed: April 25, 2026].
- [2] W. Fang et al., "Abnormal regional homogeneity in patients with essential tremor revealed by resting-state functional MRI," *PLoS One*, vol. 8, no. 7, p. e69199, Jul. 2013, doi: 10.1371/journal.pone.0069199.
- [3] E. Younger et al., "Mapping Essential Tremor to a Common Brain Network Using Functional Connectivity Analysis," *Neurology*, vol. 101, no. 15, pp. e1483–e1494, Oct. 2023, doi: 10.1212/WNL.0000000000207701.
- [4] J. Benito-León et al., "Altered Functional Connectivity in Essential Tremor: A Resting-State fMRI Study," *Medicine*, vol. 94, no. 49, p. e1936, Dec. 2015, doi: 10.1097/MD.0000000000001936.
- [5] BrainSuite Developers, "BrainSuite: Tools for the analysis of magnetic resonance images," BrainSuite. [Online]. Available: <https://brainsuite.org/>. [Accessed: Apr. 25, 2026].
- [6] NiBabel Developers, "NiBabel: Access a wide range of neuroimaging file formats," NiBabel Documentation. [Online]. Available: <https://nipy.org/nibabel/>. [Accessed: Apr. 25, 2026].
- [7] DIPY Contributors, "DIPY: Diffusion imaging in Python," DIPY Documentation. [Online]. Available: <https://dipy.org/>. [Accessed: Apr. 25, 2026].
- [8] Nilearn Developers, "Nilearn: Statistical learning for neuroimaging in Python," Nilearn Documentation. [Online]. Available: <https://nilearn.github.io/stable/index.html>. [Accessed: Apr. 25, 2026].
- [9] S. Patel, "auto_brain_connectivity: Automated analysis of neural connectivity matrices," GitHub Repository. [Online]. Available: https://github.com/sajjy/auto_brain_connectivity. [Accessed: Apr. 25, 2026].
- [10] J. Diedrichsen, "Diedrichsen 2009 cerebellar atlas," DiedrichsenLab GitHub Repository. [Online]. Available: https://github.com/DiedrichsenLab/cerebellar_atlases/tree/master/Diedrichsen_2009. [Accessed: Apr. 25, 2026].
- [11] Diedrichsen, J. (2006). A spatially unbiased atlas template of the human cerebellum. *Neuroimage*, 33, 1, p. 127-138.
- [12] Diedrichsen, J., Balsters, J. H., Flavell, J., Cussans, E., & Ramnani, N. (2009). A probabilistic atlas of the human cerebellum. *Neuroimage*.
- [13] Diedrichsen, J., Maderwald, S., Kuper, M., Thurling, M., Rabe, K., Gizewski, E. R., et al. (2011). Imaging the deep cerebellar nuclei: A probabilistic atlas and normalization procedure. *Neuroimage*.
- [14] Diedrichsen, J. & Zotow, E. (2015). Surface-based display of volume-averaged cerebellar data. *PLoS One*, 7, e0133402.